Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “voltage” to explain your model or result. Sentence starters are optional.

power: rate of energy transfer

voltage: electric potential difference; energy transferred per charge

current: rate of charge flow

01 Warm-up: bring back an earlier idea

Parallel branches draw 0.3 A and 0.5 A.
Find the current supplied to their junction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

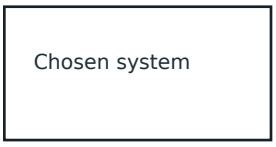
Reference: $P = IV$; $E = P\Delta t$; $P = I^2R = V^2/R$.
Constant power for $E = P\Delta t$; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power $VI=12$ W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two motors use the same energy but one takes half the time.
Compare average power.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

04 Apply the physics

A model-car motor draws 2 A at 6 V for 5 s.
Calculate input power and energy.
Label the energy-flow model with voltage, current, duration and total input, using the correct units for each.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A wave travels at fixed speed. A second target is 4 times farther away than the first.
Compare one-way travel times.
Label a wave sketch to support the calculation or comparison.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both … . They differ in … because … .



Time or position: choose and label

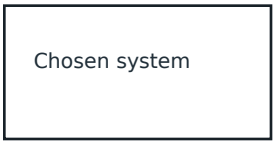
Marzano check - circle: 1 2 3

07 Exit check

A 3 W device runs for 10 s.
Find energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

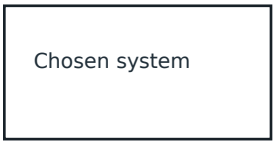
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two devices use the same 60 J: one runs at 10 W, another at 20 W.
Find their run times.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3